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Unit ⇒ 1
Electrostatics

* Vector Algebra.

Vector Algebra having magnitude as well as direction.

representation $\vec{A} = A\hat{a}_A$

A = magnitude of $\vec{A}$

$\hat{a}_A$ = unit vector along $\vec{A}$

$\left\{ \hat{a} = \frac{\vec{a}}{|\vec{a}|} \right.$

i) Dot product :-

Dot product also a scalar product. It has only magnitude.

$$\boxed{\vec{A}\cdot\vec{B} = |\vec{A}||\vec{B}|\cos\theta.}$$

ii) Cross product :-

Cross product also a vector product. It has magnitude as well as Direction both.

$$\boxed{\vec{A}\times\vec{B} = |\vec{A}||\vec{B}|\sin\theta.}$$

* Co-ordinate System :-

Orthogonal (1) Non-Orthogonal ($\neq 1$)

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i)) orthogonal.

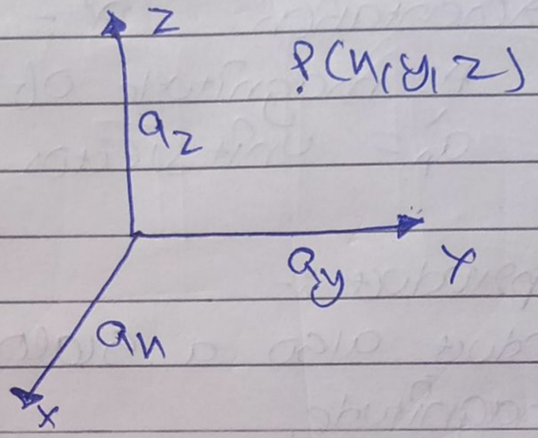
a.)) cartesion co-ordinate system (x, y, z)

b.)) cylindrical co-ordinate system (ρ, ϕ, z)

c.)) spherical co-ordinate system (r, θ, ϕ)

a.)) cartesion co-ordinate system.

Having all three linear vector.



$$P = P_x a_x + P_y a_y + P_z a_z$$

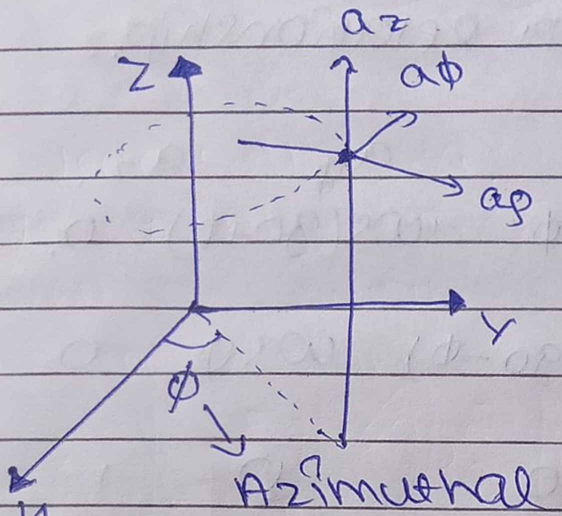
$$|\vec{P}| = \sqrt{P_x^2 + P_y^2 + P_z^2}$$

$$\boxed{a_x \cdot a_x = a_y \cdot a_y = a_z \cdot a_z = 1}$$

$$-\infty < x < \infty, \quad -\infty < y < \infty, \quad -\infty < z < \infty.$$

$a_x \times a_y = a_z$	$a_y \times a_x = -a_z$
$a_y \times a_z = a_x$	$a_z \times a_y = -a_x$
$a_z \times a_x = a_y$	$a_x \times a_z = -a_y$

b.)) Cylindrical co-ordinate System (P, Φ, Z)
Having two Linear and one angular vector.



ρ = radius of cylinder.
Φ = Azimuthal angles

Azimuthal angle {when the line is drawn ⊥ to x-y plane & there is angle with the x-axis called Azimuthal angle}.

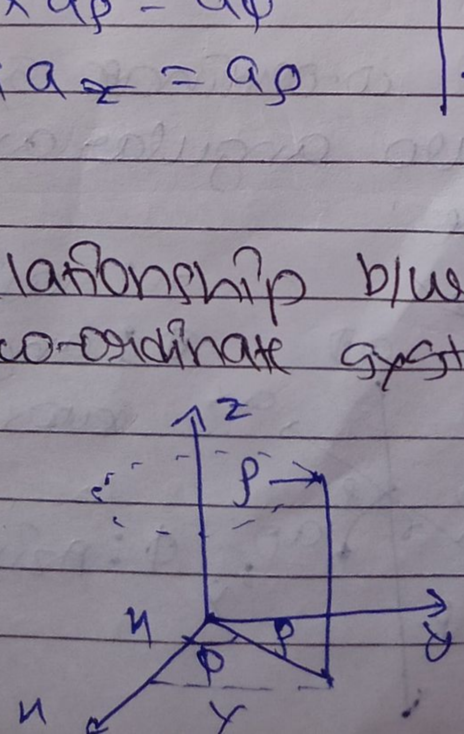
$$P = P_\rho a_\rho + P_\phi a_\phi + P_z a_z$$

$$|\vec{P}| = \sqrt{P_\rho^2 + P_\phi^2 + P_z^2}$$

$a_\rho \times a_\phi = a_z$
$a_z \times a_\rho = a_\phi$
$a_\phi \times a_z = a_\rho$

$0 < \rho < \infty$.
$0 < \phi < 2\pi$
$-\infty < z < \infty$.

* Relationship b/w cartesian & cylindrical co-ordinate system.

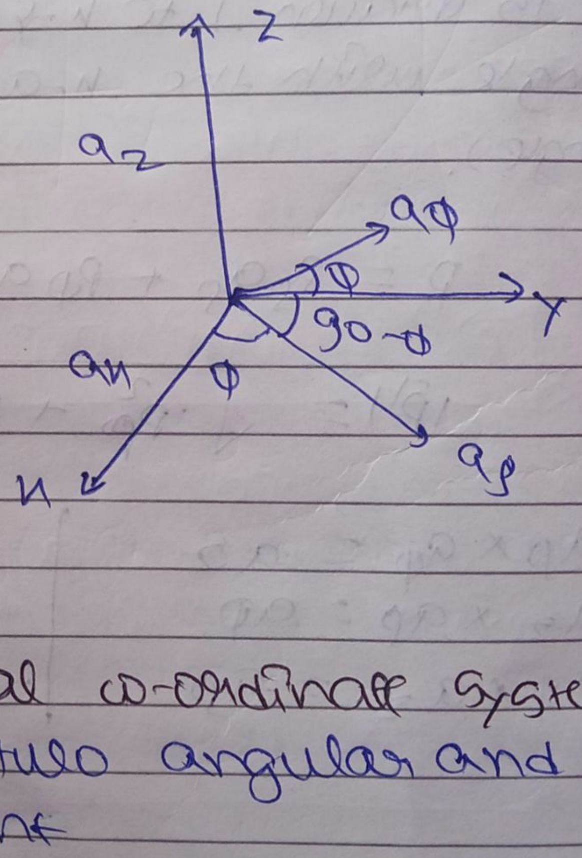


$x = \rho \cos\phi$
$y = \rho \sin\phi$
$z = z$

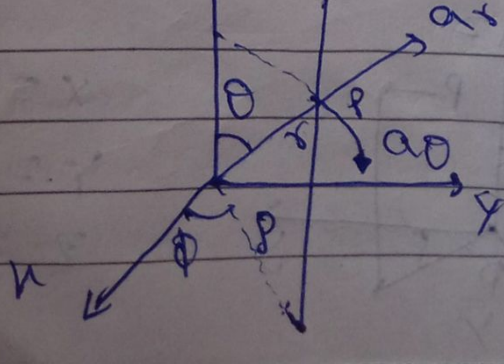
$$x^2 + y^2 = \rho^2$$
$$\phi = \tan^{-1}\left(\frac{y}{x}\right)$$

Unit vector Relationship

	a_ρ	a_ϕ	a_z
a_x	$\cos\phi$	$\cos(90+\phi)$	0
a_y	$\cos(90-\phi)$	$\cos\phi$	0
a_z	0	0	1

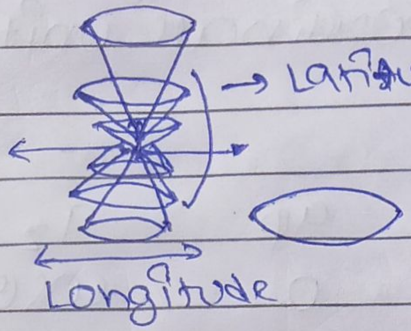


(ii) Spherical co-ordinate system.
Having two angular and one linear movement



r : radius (positive vector)
θ : angle of position vector with z-axis

ϕ : Azimuthal angle



$0 < r < \infty$
$0 < \theta < \pi$
$0 < \phi < 2\pi$

$a_\theta \times a_\phi = a_r$
$a_r \times a_\phi = -a_\theta$

Relationship b/w cartesion, cylindrical and spherical co-ordinate system,

$$\left.\begin{array}{l} x = \rho\cos\phi = r\sin\theta\cos\phi \\ y = \rho\sin\phi = r\sin\theta\sin\phi \\ z = r\cos\theta = z \end{array}\right\} \text{cartesion \& cylindrical}$$

$$\left.\begin{array}{l} \rho = r\sin\theta \\ \phi = \phi \\ z = r\cos\theta \end{array}\right\} \text{cylindrical \& spherical}$$

$$\left.\begin{array}{l} z = r\cos\theta \\ r = \sqrt{x^2+y^2+z^2} \\ \phi = \tan^{-1}\left(\frac{y}{x}\right) \\ \theta = \tan^{-1}\left(\frac{\sqrt{x^2+y^2}}{z}\right) \end{array}\right\} \text{spherical \& cartesion}$$

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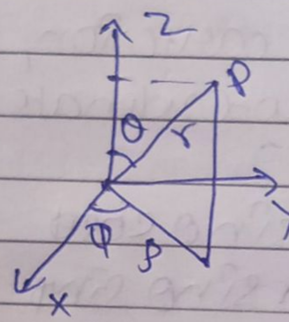
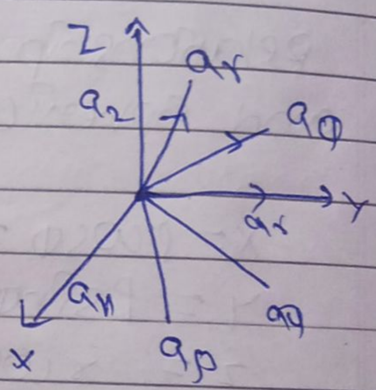
* Spherical co-ordinate unit vector relationship.

	a_ρ	a_ϕ	a_z
a_r	$\sin\theta$	0	$\cos\theta$.
a_θ	$\sin(90+\theta)$ $=\cos\theta$	0	$\cos(90+\theta)$ $-\sin\theta$
a_ϕ	0	1	0





	a_x	a_y	a_z
a_r $(a_r a_\rho)(a_\rho \cdot a_x)$		$\sin\theta\sin\phi$	$\cos\theta$
a_θ	$\cos\theta\cos\phi$	$\cos\theta\sin\phi$	$\sin\theta$
$\cdot a_\phi$	$-\sin\theta$	$\cos\phi$	0

Vector ~~calcus~~. calculas

i)) Differential length * Integral length.
ii)) Differential surface * Integral surface
iii)) Differential volume * Integral volume.

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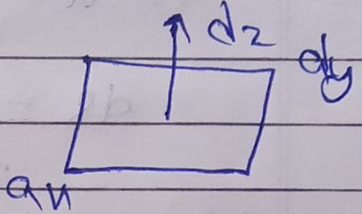
* cartesion co-ordinate system (x, y, z)

i) Different length (vector)

$$dl = d_x a_x + d_y a_y + d_z a_z$$

ii) different surface (vector).

$$ds = dx\,dy\,a_z$$
$$= dx\,dz\,a_y$$
$$= dy\,dz\,a_x$$



iii) Different volume (scalar).

$$dv = dx\,dy\,dz$$

* cylindrical system (ρ, ϕ, z).

i) different length.

$$dl = d\rho\, a_\rho + \rho\, d\phi\, a_\phi + d_z\, a_z$$

ii) different surface.

$$ds = d\rho \cdot d_z\, a_\phi$$
$$= \rho\, d\phi\, dz\, a_\rho$$
$$= \rho\, d\rho \cdot d\phi\, a_z$$

iii) different volume.

$$dv = \rho\, d\rho\, d\phi\, dz$$

* Spherical co-ordinate system (r, θ, ϕ)

i)) different length.

$$dl = dr a_r + r d\theta a_\theta + r \sin\theta d\phi a_\phi$$

ii)) different surface.

$$ds = r d\theta . r \sin\theta d\phi a_r$$
$$= r dr \sin\theta d\phi a_\theta$$
$$= r dr d\theta a_\phi$$

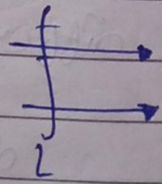
iii)) different volume.

$$dv = r^2 \sin\theta dr d\theta d\phi$$

Integrals

i)) Line Integral

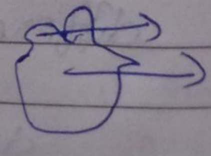
$$\int A \cdot dl = \int A dl \cos\theta$$



circulation of A around L

ii)) Surface Integral.

$$\int A \cdot ds = \int A ds \cos\theta$$



Net Flux Through surface

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(iii) Volume Integral.

$$\int \rho v \, dv = \iiint \rho v \, dv.$$

Del operator ($\vec{\nabla}$).

$$\vec{\nabla} = \frac{\partial}{\partial x}\hat{x} + \frac{\partial}{\partial y}\hat{y} + \frac{\partial}{\partial z}\hat{z}$$

or

$$\vec{\nabla} = \frac{\partial}{\partial x}a_x + \frac{\partial}{\partial y}a_y + \frac{\partial}{\partial z}a_z$$

Divergence of vector $\vec{A}$

$$\vec{\nabla}\cdot\vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}.$$

$$\vec{A} = A_x\cdot\hat{x} + A_y\cdot\hat{y} + A_z\cdot\hat{z}.$$

Or

$$\vec{A} = A_x\cdot a_x + A_y a_y + A_z a_z.$$

$$\vec{\nabla} = \frac{\partial}{\partial x}\hat{x} + \frac{\partial}{\partial y}\hat{y} + \frac{\partial}{\partial z}\hat{z}.$$

Or

$$\vec{\nabla} = \frac{\partial}{\partial x}a_x + \frac{\partial}{\partial y}a_y + \frac{\partial}{\partial z}a_z.$$

$$\boxed{\vec{\nabla}\cdot\vec{A} = \frac{\partial}{\partial x}A_x + \frac{\partial}{\partial y}A_y + \frac{\partial}{\partial z}A_z}$$

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Curl of vector.

$$\vec{A} = A_x \hat{x} + A_y \hat{y} + A_z \hat{z}$$

OR

$$\vec{A} = A_x a_x + A_y a_y + A_z a_z$$

$$\vec{\nabla} = \frac{\partial}{\partial x}\hat{x} + \frac{\partial}{\partial y}\hat{y} + \frac{\partial}{\partial z}\hat{z}$$

OR

$$\vec{\nabla} = \frac{\partial}{\partial x}a_x + \frac{\partial}{\partial y}a_y + \frac{\partial}{\partial z}a_z$$

$$\vec{\nabla}\times\vec{A} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

* Gradient of ~~vector~~ scalar (V)

Gradient of scalar is vector quantity. It consist of magnitude as well as direction.

The Direction of Gradient is along the normal to the given function. Gradient measure slope

$$dv = \frac{\partial v}{\partial x}dx + \frac{\partial v}{\partial y}dy + \frac{\partial v}{\partial z}dz$$

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$$dv = \underbrace{\left(\frac{\partial v}{\partial x} a_x + \frac{\partial v}{\partial y} a_y + \frac{\partial v}{\partial z} a_z\right)}_{\text{Gradient.}} \left(dx a_x + dy a_y + dz a_z\right)$$

$dv = G \cdot dl.$

$dv = G\,dl \cos\theta.$

$$\boxed{G\cos\theta = \frac{dv}{dl}}$$ if $\theta = 0$

$$\boxed{G = \frac{dv}{dl}}$$

Q. Find Gradient of r^2, where r is position vector.

$\vec{r} = x a_x + y a_y + z a_z.$

$|\vec{r}| = \sqrt{x^2+y^2+z^2}$

$|\vec{r}|^2 = x^2+y^2+z^2.$

$\vec{\nabla}^2 = \vec{\nabla}(x^2+y^2+z^2)$

$= \left(\frac{\partial}{\partial x}\hat{x} + \frac{\partial}{\partial y}\hat{y} + \frac{\partial}{\partial z}\hat{z}\right)(x^2+y^2+z^2)$

$= 2x\hat{x} + 2y\hat{y} + 2z\hat{z}.$ Any

* Divergence of vector ($\vec{A}$).
Divergence of any vector is scalar quantity. Divergence of vector ($\vec{A}$) at any point is net outward flux per unit volume.

$\nabla \cdot A$ = Divergence of vector.

Q Calculate Divergence of vector $\vec{r}$ where $\vec{r}$ is position vector.

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}.$$

Divergence is $\vec{\nabla} \cdot \vec{r}$

$$\vec{\nabla} \cdot \vec{r} = \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right)\left(x\hat{i} + y\hat{j} + z\hat{k}\right).$$

$$\vec{\nabla} \cdot \vec{r} = \frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial z}{\partial z}$$

$$\vec{\nabla} \cdot \vec{r} = 1 + 1 + 1$$

$$\boxed{\vec{\nabla} \cdot \vec{r} = 3}$$

Scalar Quantity.

* Note :- If Divergence is zero then it is called solenoidal vector.

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* Divergence Theorem or Gauss Theorem,

According to this Theorem, The Divergence of any vector over the volume Integral is equal to the surface Integral of same vector.

$$\boxed{\iiint (\nabla \cdot \vec{A})\, dv = \iint \vec{A} \cdot \vec{ds}}$$

Curl of a vector ($\vec{A}$).

Curl of vector is a vector quantity. The curl of a vector is a cross product between Del ($\vec{\nabla}$) and Any Given vector.

$$(\vec{\nabla} \times \vec{A}) = \text{Curl of vector.}$$

Where,

$$\vec{\nabla} = \frac{\partial}{\partial x}\hat{x} + \frac{\partial}{\partial y}\hat{y} + \frac{\partial}{\partial z}\hat{z}.$$

$$\vec{A} = A_x\hat{x} + A_y\hat{y} + A_z\hat{z}.$$

$$\vec{\nabla} \times \vec{A} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

Q. Find the curl of vector $\vec{r}$ where $\vec{r}$ is position vector.

Sol
$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}.$$

$$\vec{\nabla} = \frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}.$$

$$\vec{\nabla}\times\vec{r} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & y & z \end{vmatrix}$$

$$\vec{\nabla}\times\vec{r} = \hat{i}\left(\frac{\partial z}{\partial y} - \frac{\partial y}{\partial z}\right) - \hat{j}\left(\frac{\partial z}{\partial x} - \frac{\partial x}{\partial z}\right) + \hat{k}\left(\frac{\partial y}{\partial x} - \frac{\partial x}{\partial y}\right)$$

$$\vec{\nabla}\times\vec{r} = 0 + 0 + 0 = 0.$$

Note :-
If curl of any vector is zero $\vec{\nabla}\times\vec{A} = 0$ then it is known as Irrotational vector.
It means there is no rotation

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* Stokes theorem.

According to this theorem, the curl of any vector over surface Integral is equal to the Line Integral of same vector.

$$\boxed{\iint (\vec{\nabla} \times \vec{A}) \cdot d\vec{s} = \int \vec{A} \cdot d\vec{l}}$$

Coulomb's Law for electrostatic.

According to coulomb's law, the force b/w two point charge is directly proportional to the product of magnitude of two point charge and inversely proportional to ~~to~~ square of radius (Inverse square law).

$$q_1 \text{ - - - - - - - - - - - } q_2$$
$$\longleftarrow \; r \; \longrightarrow$$

$$F \propto |q_1||q_2| \quad \text{--- (1)}$$

$$F \propto \frac{1}{r^2} \quad \text{--- (2)}$$

From equation (1) & (2)

$$F \propto \frac{q_1 q_2}{r^2}$$

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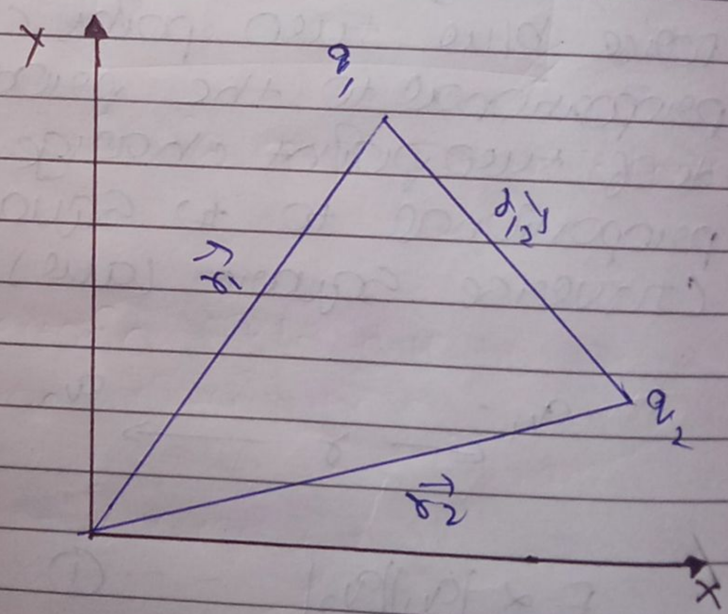
$$\boxed{F = \frac{k q_1 q_2}{r^2}}$$

where

$$k = \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \, Nm^2/C^2$$

$$\epsilon_0 = 8.85 \times 10^{-12} \, C^2/Nm^2$$

* vector form of coulomb's law.



$$F = \frac{k q_1 q_2}{r^2}$$

$$\vec{F_{21}} = \frac{k q_1 q_2}{r_{12}^2} \times \hat{r}_{12}$$

$$\vec{F_{21}} = \frac{K q_1 q_2}{|\vec{r_{12}}^2|} \times \frac{\vec{r_{12}}}{|\vec{r_{12}}|}$$

$$\vec{F_{21}} = \frac{K q_1 q_2 \vec{r_{12}}}{|r_{12}|^3} \quad \text{--- (1)}$$

Now from vector law of addition

$$\vec{r_1} + \vec{r_{12}} = \vec{r_2}$$

$$\vec{r_{12}} = \vec{r_2} - \vec{r_1}$$

put the value of $\vec{r_{12}}$ in eq (1)

Then,

$$\vec{F_{21}} = \frac{K q_1 q_2}{|\vec{r_2} - \vec{r_1}|^3} (\vec{r_2} - \vec{r_1}) \quad \text{- (2)}$$

My.

$$\vec{F_{12}} = \frac{K q_1 q_2}{|\vec{r_1} - \vec{r_2}|^3} (\vec{r_1} - \vec{r_2}) \quad \text{-- (3)}$$

So from eq (2) & (3)

$$\boxed{\vec{F_{12}} = -\vec{F_{21}}} \rightarrow$$ It follow newton 3rd law of motion

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* Electrostatic field.
The electrostatic field is defined as the electrostatic force per unit charge (q_0).

$$\text{Electric field} = \frac{\text{Force}}{\text{charge}}$$

$$E = \frac{kq\cancel{q_0}}{r^2\cancel{q_0}}$$

$$\boxed{E = \frac{kq}{r^2}}$$

S.I unit is N/C or volt/metre.

* Electric field due to continuous charge distribution.

i) point charge 'Q'

ii) line charge or linear charge density

$$\lambda = \frac{\text{charge }(q)}{\text{Length }(l)}$$

$\boxed{q = \lambda l}$ or $q = \int_L \lambda_L \, dl$

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ii) Surface charge Density (σ)

$$\sigma = \frac{q}{A} = \frac{\text{charge}}{\text{Area}}$$

$$\boxed{q = \sigma A} \Rightarrow q = \int_S \rho_s \, ds$$

iii) volume charge Density (ρ).

$$\rho = \frac{\text{charge}}{\text{volume}} = \frac{q}{v}.$$

$$\boxed{q = \rho v.} \Rightarrow q = \int_V \rho_v \, dv$$

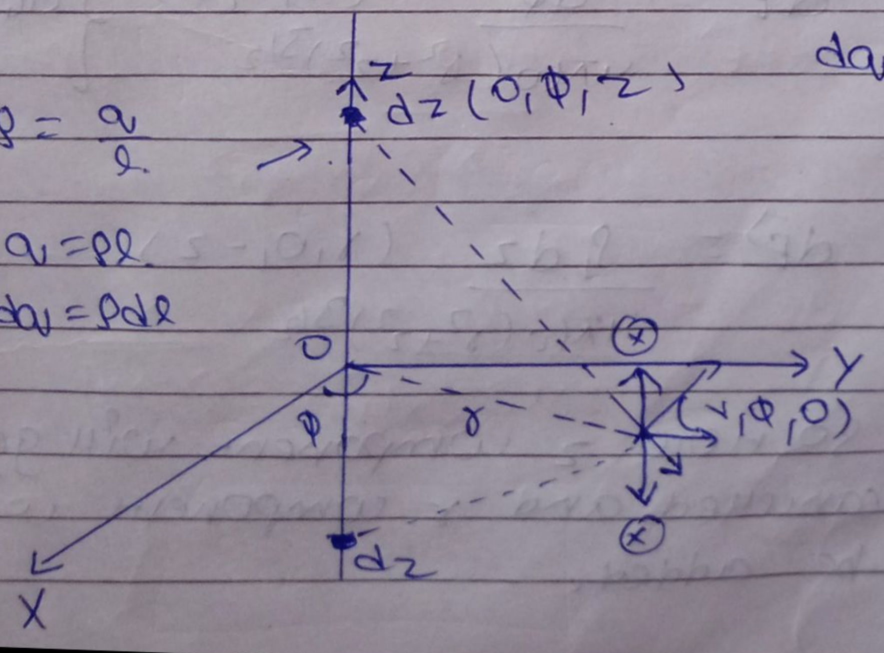
* calculation Electric Field Intensity due to line charge.

$\rho = \frac{q}{l}$

$q = \rho l.$

$dq = \rho dl$

$dq = \rho dz.$



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$$\vec{E} = \frac{KQ}{R^2} \cdot \hat{R}$$

$$d\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{dQ}{R^2} \cdot \frac{\vec{R}}{|\vec{R}|}$$

$$d\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{dQ}{R^3} \cdot \vec{R} \quad \text{--- (1)}$$

Direction of $\vec{R}$.

$$\vec{R} = (r, \phi, 0) - (0, \phi, z)$$

$$\vec{R} = (r, 0, -z)$$

magnitude,

$$|\vec{R}| = \sqrt{r^2 + z^2}$$

put the magnitude of $|\vec{R}|$ in eq(1).

$$d\vec{E} = \frac{dQ}{4\pi\epsilon_0 (r^2+z^2)^{3/2}} \cdot (r, 0, -z) \quad \left\{ dQ = \rho_L dz \right\}$$

$$d\vec{E} = \frac{\rho_L dz}{4\pi\epsilon_0 (r^2+z^2)^{3/2}} (r, 0, -z)$$

So here z component will get cancelled and r component will be added.

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$$d\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{\rho_L dz}{(r^2+z^2)^{3/2}} (r,0,0).$$

Integrate both side.

$$\vec{E} = \int_{-\alpha}^{\alpha} \frac{1}{4\pi\epsilon_0} \frac{\rho_L dz}{(r^2+z^2)^{3/2}} (r,0,0).$$

Now,

$z = r\tan\theta.$

$dz = r\sec^2\theta d\theta.$

Limit $-\alpha$ to α.

$-\alpha \rightarrow -\frac{\pi}{2}, \quad \alpha \rightarrow \frac{\pi}{2}$

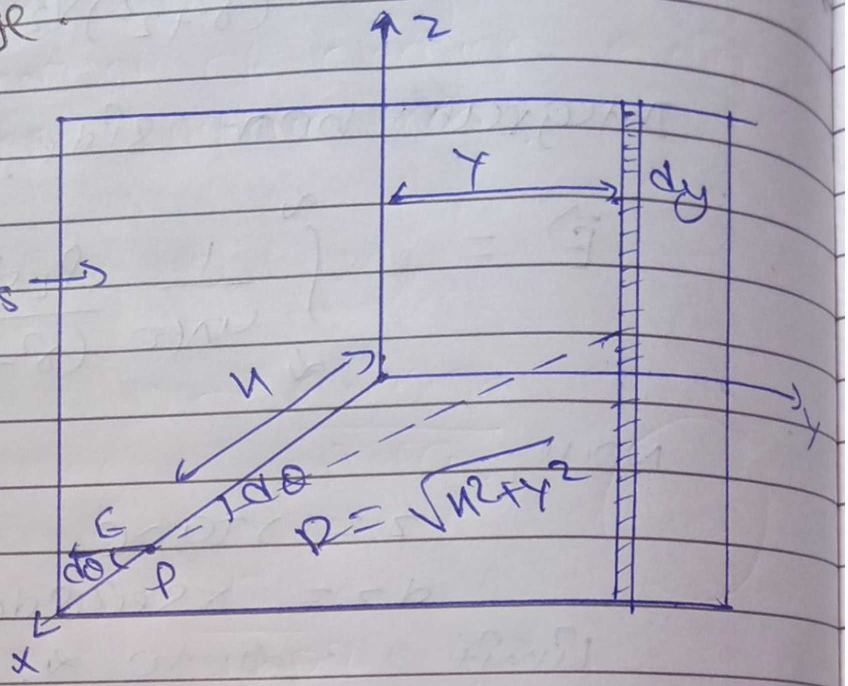
$$\vec{E} = \int_{-\frac{\pi}{2}}^{\pi/2} \frac{1}{4\pi\epsilon_0} \frac{\rho_L \times r\sec^2\theta d\theta}{[r^2+r^2\tan^2\theta]^{3/2}} \hat{a}_r$$

$$\vec{E} = \frac{\rho_L}{4\pi\epsilon_0} \int_{-\frac{\pi}{2}}^{\pi/2} \frac{\cancel{r^2}\sec^{\cancel{2}}\theta d\theta}{r^3\sec^3\theta} \hat{a}_r$$

$$\vec{E} = \frac{\rho_L}{4\pi\epsilon_0 r} \int_{-\frac{\pi}{2}}^{\pi/2} \cos\theta d\theta \ \hat{a}_r$$

$$\vec{E} = \frac{\rho_L}{\cancel{4}\pi\epsilon_0 r} \times \cancel{2} \hat{a}_r \Rightarrow \boxed{\vec{E} = \frac{\rho_L}{2\pi\epsilon_0 r} \hat{a}_r}$$

* Electric field intensity due to surface charge.



$\cos\theta = \frac{x}{R}$

$\cos\theta = \frac{x}{\sqrt{x^2+y^2}}$

There is no y & z component of electric field.

For line charge electric field,

$$d\vec{E} = \frac{\rho_L}{2\pi\epsilon_0 R} \qquad \left\{ \rho_s = \frac{\rho_L}{l} \quad \rho_L = \rho_s \, dy \right\}$$

so,

$$d\vec{E} = \frac{dy\,\rho_s \cos\theta}{2\pi\epsilon_0 \sqrt{x^2+y^2}} \qquad \left\{ R = \sqrt{x^2+y^2} \right\}$$

$$d\vec{E} = \frac{dy\,\rho_s}{2\pi\epsilon_0 \sqrt{x^2+y^2}} \times \frac{x}{\sqrt{x^2+y^2}}$$

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$$d\vec{E} = \frac{dy\, \rho_s\, x}{2\pi\epsilon_0 (x^2+y^2)}$$

Integrate both side.

$$\vec{E} = \int_{-\infty}^{\infty} \frac{dy\, \rho_s\, x}{2\pi\epsilon_0 (x^2+y^2)}$$

$$\vec{E} = \frac{\rho_s}{2\pi\epsilon_0} \int_{-\infty}^{\infty} \frac{x}{x^2+y^2}\, dy.$$

$$\vec{E} = \frac{\rho_s}{2\pi\epsilon_0} \left[\tan^{-1}\left(\frac{y}{x}\right) \right]_{-\infty}^{\infty}$$

$$\vec{E} = \frac{\rho_s}{2\pi\epsilon_0} \times \left[\frac{\pi}{2} + \frac{\pi}{2} \right]$$

$$\vec{E} = \frac{\rho_s}{2\cancel{\pi}\epsilon_0} \times \cancel{\pi}$$

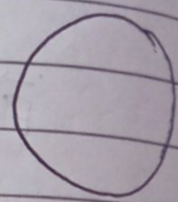
$$\boxed{\vec{E} = \frac{\rho_s}{2\epsilon_0}}$$

Electric flux Density.

Electric flux Density is defined as the amount of charge is displaced per unit surface Area.

According to flux,

$$D = \frac{\text{Charge}}{\text{Area}}$$



$$D = \frac{Q}{4\pi R^2}$$

The Direction of Electric flux is along the direction of Electric field.

$$\boxed{\vec{D} = \frac{Q}{4\pi R^2}\,\hat{R}.}$$

S.I unit of Electric flux Density is coulomb/metre^2 (C/m^2)

$$\vec{D} = \frac{\text{charge}}{\text{Area}} = \frac{\text{coulomb}}{m^2}$$

S.I unit is C/m^2.

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* Relationship between Electric Flux Density & Electric Field.

Electric Flux Density.

$$\vec{D} = \frac{Q}{4\pi R^2} \hat{R} \quad \text{--- (1)}$$

Electric Field

$$\vec{E} = \frac{KQ}{R^2} = \frac{1}{4\pi\epsilon_0} \times \frac{Q}{R^2}.$$

From equation (1).

$$\vec{E} = \frac{1}{\epsilon_0} \times \frac{Q}{4\pi R^2} \hat{R}$$

$$\vec{E} = \frac{\vec{D}}{\epsilon_0}$$

$$\boxed{\vec{D} = \epsilon_0 \vec{E}.}$$

Where,

D = Electric Flux Density.

ϵ_0 = Epsilon knot (8.85×10^{-12} Nm²/C²)

E = Electric Field.

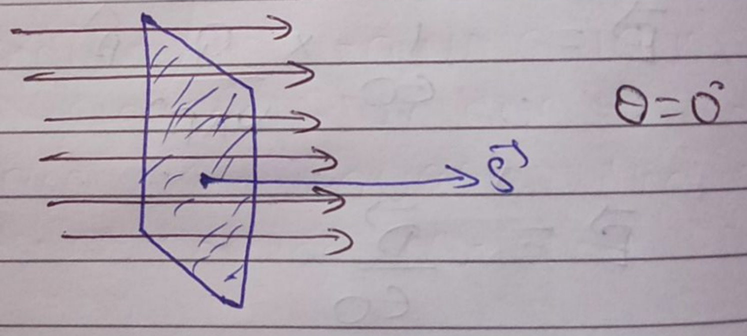
In General,
For free space,

$$\vec{D} = \epsilon \vec{E} \qquad \left\{ \epsilon_r = \frac{\epsilon}{\epsilon_0} \right.$$

$$\boxed{\vec{D} = \epsilon_r \epsilon_0 \vec{E}} \qquad \boxed{\epsilon = \epsilon_r \epsilon_0}$$

* Electric Flux.

The total number of Electric field passing through the given surface. Numerically it is equal to the product of Electric field and surface Area.



Electric flux = Electric field × Area

$$\psi = \vec{E} \cdot d\vec{A}$$

$$\boxed{\psi = \vec{E}\, d\vec{A} \cos\theta}$$

Here $\theta = 0^\circ$

So,

$$\psi = \vec{E}\, d\vec{A} \cos 0^\circ$$

$$\boxed{\begin{gathered}\psi = \vec{E}\, d\vec{A} \\ \text{Or} \\ \psi = ES\end{gathered}}$$

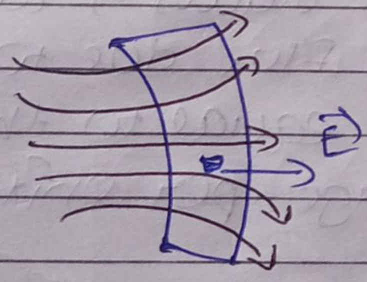
where
S = Surface Area.

S.I unit of Electric Flux.

$$\psi = \vec{E} \times \vec{A}$$

$$\psi = \frac{\text{volt}}{\cancel{\text{metre}}} \times m^{\cancel{2}}$$

S.I unit of Electric Flux is ~~V/m~~ volt-metre (v-m) or waber (wb).



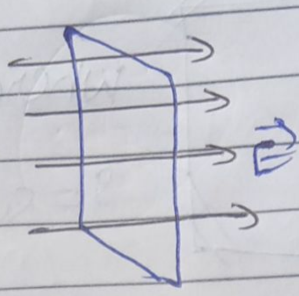
Non-uniform.

$$d\psi = \vec{E} \cdot d\vec{S}$$

Integrate both side.

$$\boxed{\psi = \int \vec{E} \cdot d\vec{S}}$$

Case I :-

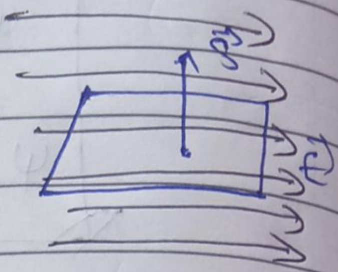


$\psi = \int E \cdot dS \cos\theta$

$\theta = 0.$

$\boxed{\psi = ES.}$

Case II :



$\psi = \vec{E} \vec{dS} \cos\theta$

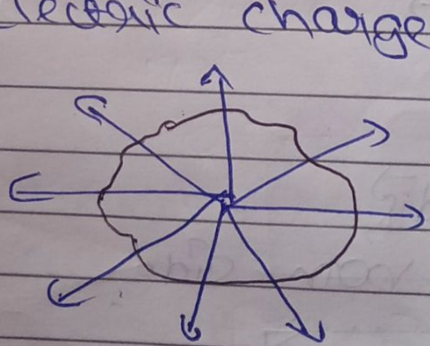
$\theta = 90°$

$\psi = \vec{E} \vec{dS} \cos 90°$

$\boxed{\psi = 0}$

Gauss's Law for Electric Field

According to Gauss's Law, the total electric flux due to any closed surface is equal to the net enclosed electric charge per unit epsilon not.



Electric Flux = $\frac{\text{Charge (enclosed)}}{\text{Epsilon not}}$

$$\boxed{\psi = \frac{q_{enclosed}}{\varepsilon_0}}$$

So,

$$\boxed{\psi = \oint \vec{E} \cdot d\vec{s} = \frac{Q}{\varepsilon_0}}$$

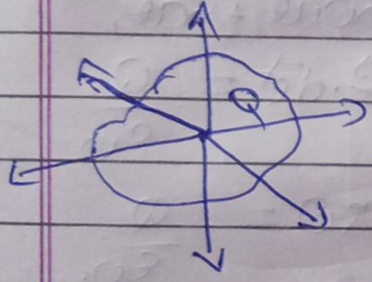
Or.

$$\boxed{\psi = \oint \vec{D} \cdot d\vec{s} = Q}$$

Where

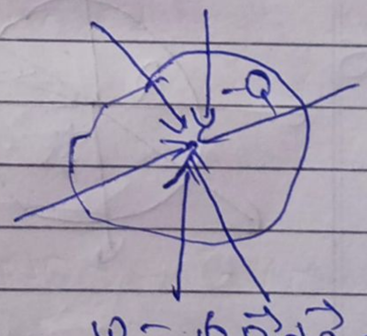
D = Electric flux density.

Case I



$\psi = \oint \vec{D} \cdot d\vec{s} = Q_{encl.}$

Case II

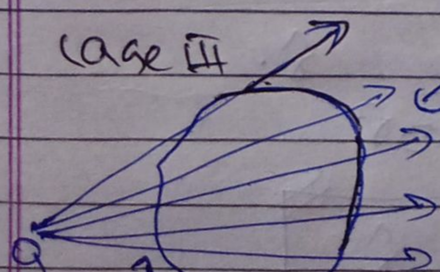


$\psi = \oint \vec{D} \cdot d\vec{s} = -Q_{encl}$

Note:

If flux is leaving enclosed surface than Electric flux is positive & if Electric flux is entering inside enclosed surface then electric flux is negative.

Case III

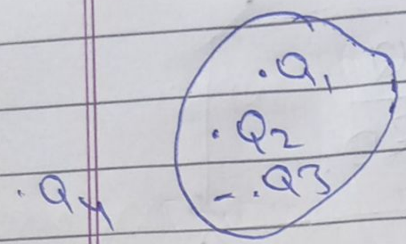


Electric flux is leaving surface. So flux is positive

So net flux is zero.

Flux is negative because it entering

Case IV



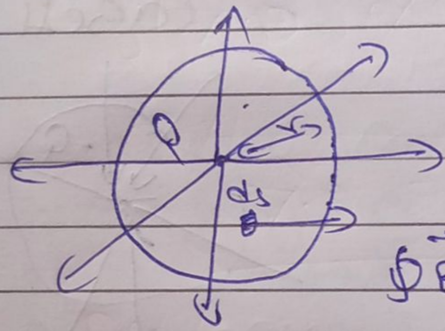
$$\varphi = \oint \vec{E}\cdot\vec{ds} = \frac{(Q_1+Q_2-Q_3)}{\varepsilon_0}$$

or

$$\varphi = \oint \vec{D}\cdot\vec{ds} = (Q_1+Q_2-Q_3)$$

Application of Gauss's law for Electric field.

i) Electric field due to point charge.



$\theta = 0^\circ$

We know that,

$$\oint \vec{E}\cdot\vec{ds} = \frac{Q}{\varepsilon_0}$$

$$\oint \vec{E}\,\vec{ds}\cos\theta = \frac{Q}{\varepsilon_0}$$

$$\oint \vec{E}\,\vec{ds}\cos 0^\circ = \frac{Q}{\varepsilon_0}$$

$$\oint_0^{4\pi r^2} \vec{E}\,\vec{ds} = \frac{Q}{\varepsilon_0}$$

$$E \times 4\pi r^2 = \frac{Q}{\varepsilon_0}$$

$$\vec{E} = \frac{Q}{4\pi\varepsilon_0 r^2}\,\hat{r}.$$

$$\boxed{\vec{E} = \frac{kQ}{r^2}\,\hat{r}}$$

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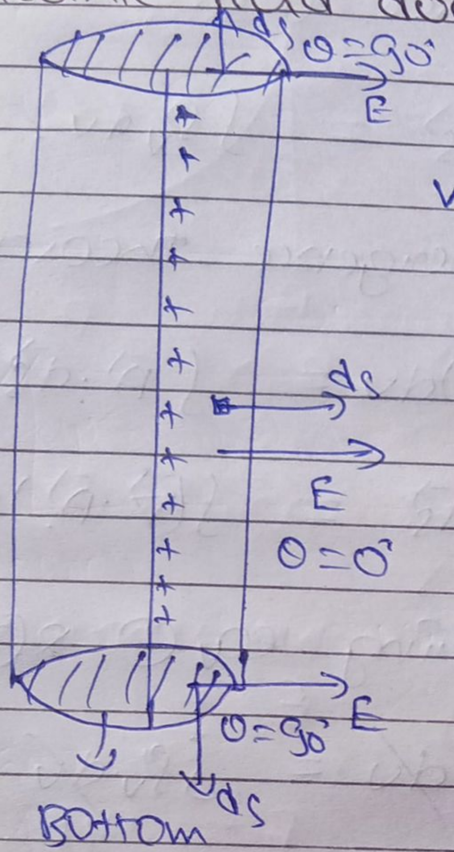
iii) Electric field due to line charge.



$\rho_l = \frac{Q}{l}$

$Q = \rho_l \cdot l$.

We know that,

$$\oint \vec{E}\cdot\vec{ds} = \frac{Q}{\varepsilon_0}$$

$$\oint E\, ds \cos 0° = \frac{Q}{\varepsilon_0}$$

$$\vec{E} \times \vec{A} \times \cancel{l} = \frac{\rho_L \cdot \cancel{l}}{\varepsilon_0}$$

$$\boxed{\vec{E} = \frac{\rho_L}{2\pi\varepsilon_0 r}\, \hat{r}}$$

Maxwell's I^st equation with integral form & differential form.

i) From Gauss law,

$$\oint \vec{D}\cdot\vec{ds} = Q \quad \text{--- (1)}$$

For volume charge Distribution

$$\rho_v = \frac{dQ}{dv} \Rightarrow Q = \int \rho_v \cdot dv.$$

Put the value of Q in eq (1)

$$\boxed{\oint \vec{D}\cdot\vec{ds} = \int \rho_v \cdot dv}$$

Integral form.

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(iii) Differential form.

$$\oint \vec{D}\cdot\vec{ds} = \int \rho_v \, dv \quad \text{--- (1)}$$

From Divergence Theorem.

$$\int (\vec{\nabla}\cdot\vec{A})\, dv = \int \vec{A}\cdot\vec{ds}$$

$$\int \vec{A}\cdot\vec{ds} = \int (\vec{\nabla}\cdot\vec{A})\, dv \quad \text{--- (2)}$$

On comparing eq (1) & (2).

$$\int (\vec{\nabla}\cdot\vec{D})\, \cancel{dv} = \int \rho_v \, \cancel{dv}$$

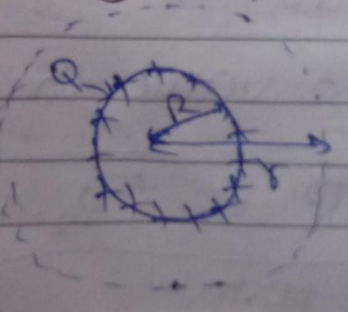
$$\boxed{\vec{\nabla}\cdot\vec{D} = \rho_v}$$

Or

$$\boxed{\vec{\nabla}\cdot\vec{E} = \frac{\rho_v}{\epsilon_0}}$$

This is Differential form of maxwell's equation (1).

* Electric Field Intensity due to conducting sphere.



Case I

$r > R$

From Gauss's law

$$\oint \vec{E} \cdot \vec{ds} = \frac{Q}{\varepsilon_0}$$

$\theta = 0^\circ$

$$\oint \vec{E}\,\vec{ds} \cos 0^\circ = \frac{Q}{\varepsilon_0} \quad \{\vec{E} \parallel \vec{ds}\}$$

$$\vec{E} \oint_0^{4\pi r^2} \vec{ds} = \frac{Q}{\varepsilon_0}$$

$$\boxed{\vec{E} = \frac{Q}{4\pi r^2 \varepsilon_0} \hat{r}}$$

Case II

$r = R$

$$\oint \vec{E} \cdot \vec{ds} = \frac{Q}{\varepsilon_0} \quad \{\theta = 0^\circ\}$$

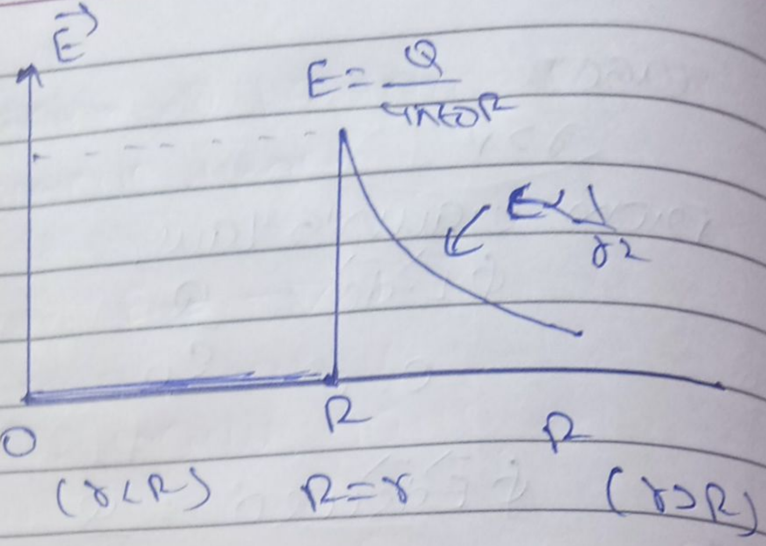
$$E \oint_0^{4\pi R^2} ds = \frac{Q}{\varepsilon_0}$$

$$\boxed{\vec{E} = \frac{Q}{4\pi R^2 \varepsilon_0} \hat{r}}$$

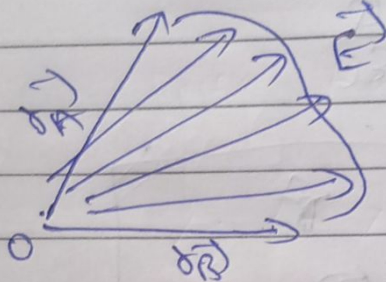
Case III $r < R$

$$\oint \vec{E} \cdot \vec{ds} = \frac{Q}{\varepsilon_0} = 0 \Rightarrow \boxed{\vec{E} = 0}$$

Inside the sphere



Electrostatic potential.



We know that,

$E = \frac{F}{q}$.

$F = qE$.

For dl movement workdone

$dw = \vec{F} \cdot \vec{dl}$ $\{F = qE\}$

$dw = q\vec{E} \cdot \vec{dl}$

Integrate both side

$$\boxed{W = \int_A^B q\vec{E} \cdot \vec{dl}}$$

$\left\{ \begin{array}{l} W = \frac{kq_1q}{r} \\ E = \frac{kq}{r^2} \\ W = qEr \end{array} \right.$

potential Difference.

$V_{AB} = \frac{W}{q} = \frac{\cancel{q}}{\cancel{q}} \int_A^B \vec{E} \cdot \vec{dl}$

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$$\boxed{V_{AB} = -\int_A^B \vec{E} \cdot \vec{dl}.}$$ $\left\{ V_{AB} = V_B - V_A \right\}$

{-ve sign is due to workdone by external force}

$$V_{AB} = -\int_A^B \frac{Kq}{r^2}\, dr.$$

$$V_{AB} = -Kq \left(-\frac{1}{r} \right)_A^B$$

$$\boxed{V_{AB} = Kq \left[\frac{1}{r_B} - \frac{1}{r_A} \right]}$$

From a point to infinity.

$$V = \frac{Kq}{r}$$

$$\boxed{V = \frac{q}{4\pi\epsilon_0 r}}$$

S.I unit is J/C or volt.

Poission & Laplace equation

We know that,

$$\vec{E} = -\vec{\nabla} V \quad --- (1)$$

From Gauss law,

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho_v}{\epsilon_0} \quad --- (2)$$

Put the value of E in eq(2)

$$\vec{\nabla} \cdot (-\vec{\nabla} V) = \frac{\rho_v}{\epsilon_0}$$

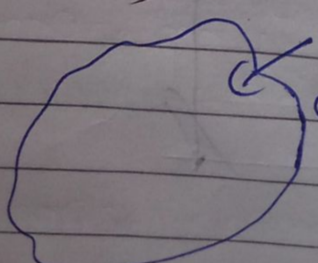
$$\vec{\nabla} \cdot (\vec{\nabla} V) = -\frac{\rho_v}{\epsilon_0}$$

$$\boxed{\nabla^2 V = -\frac{\rho_v}{\epsilon_0}}$$

{ ∇^2 = Laplace operator.

Poission equation.

Now,



If enclosed surface does not have any volu[me] charge.

so $\rho_v = 0$

$$\nabla^2 V = -\frac{\rho_v}{\epsilon_0} = 0$$

$$\boxed{\nabla^2 V = 0}$$

Laplace equation.

* Uniqueness Theorem.

Uniqueness theorem explains unique solution to Laplace equation.

From Laplace eqⁿ

$$\nabla^2 V = 0$$

There are two solution, V_1 & V_2

$$\nabla^2 V_1 = 0 \quad \& \quad \nabla^2 V_2 = 0$$

If $V_3 = V_1 - V_2$, for same boundary.

$$\nabla^2 V_3 = 0$$

For metallic boundary because of same boundary,

$$V_3 = 0 \Rightarrow V_1 - V_2 = 0$$

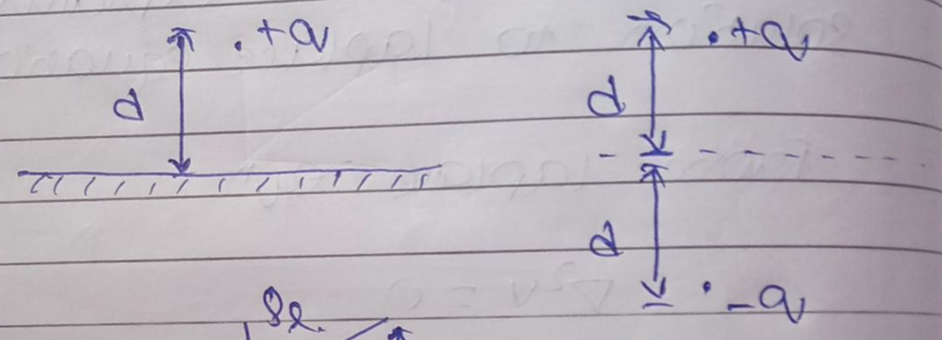
$$\boxed{V_1 = V_2}$$

There is no maxima & minima enclosed surface.

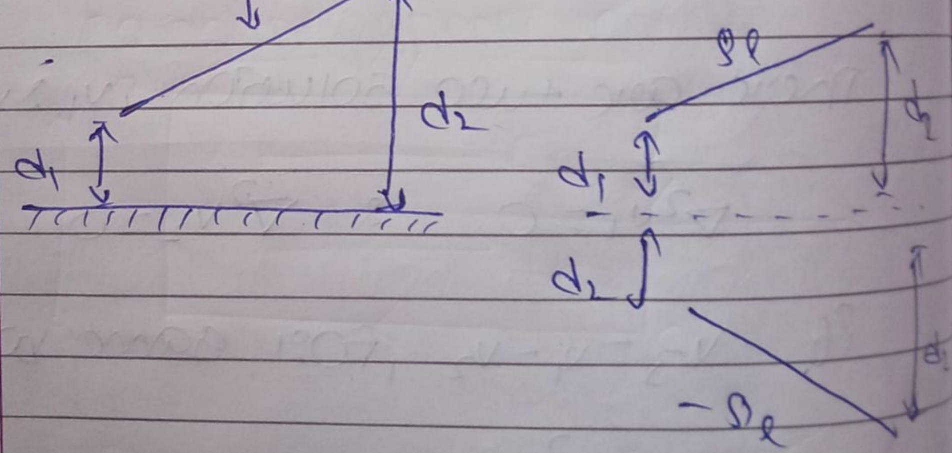
Image Theory of charges

It states that if any charge is above infinite grounded perfect conducting plane then it is replaced by charge with it's image and with opposite polarity.

i))



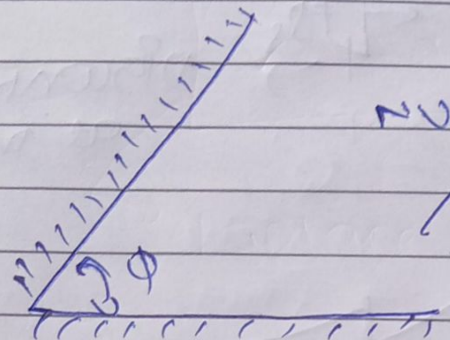
ii))



Important points

i)) Image charges must be there inside the conducting region.

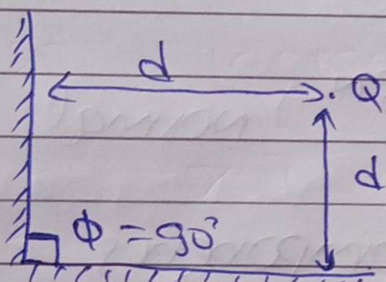
ii)) Image charge must be located in a way that on perfect conducting plane has voltage (V=0) on it's surface.



number of image charges

$$\boxed{N = \frac{360^\circ}{\phi} - 1}$$

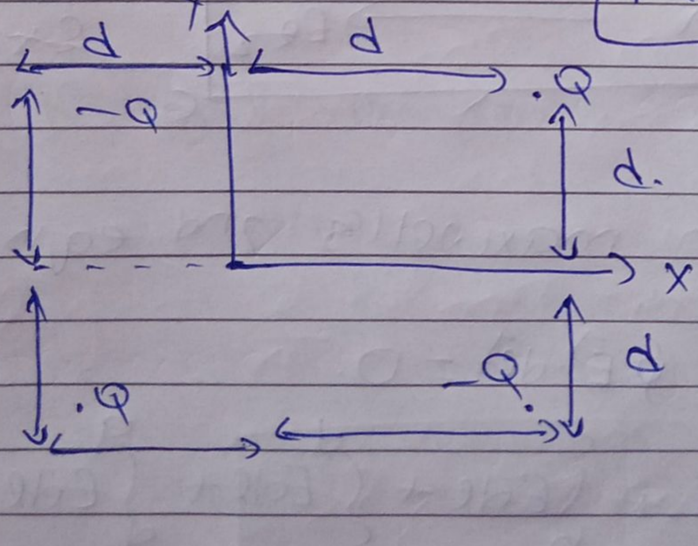
Example :-



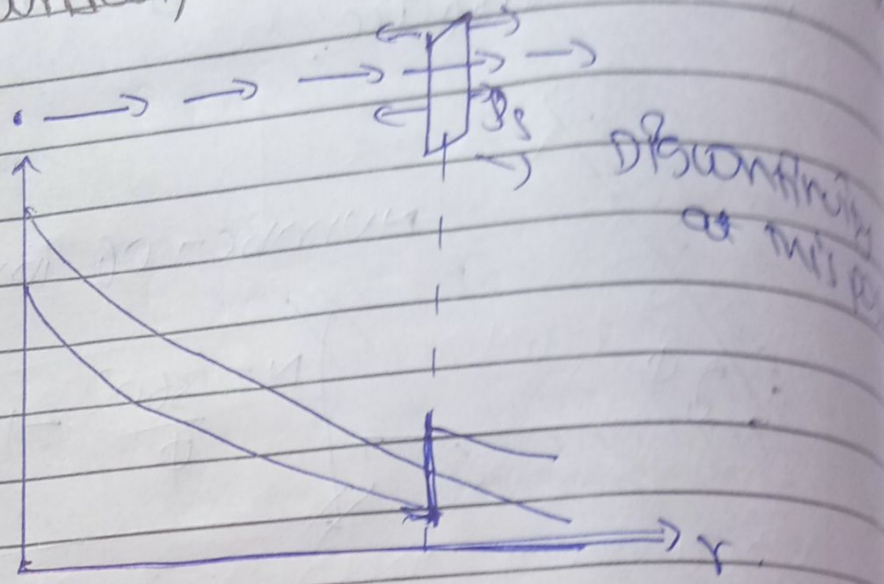
$$N = \frac{360^\circ}{\phi} - 1$$

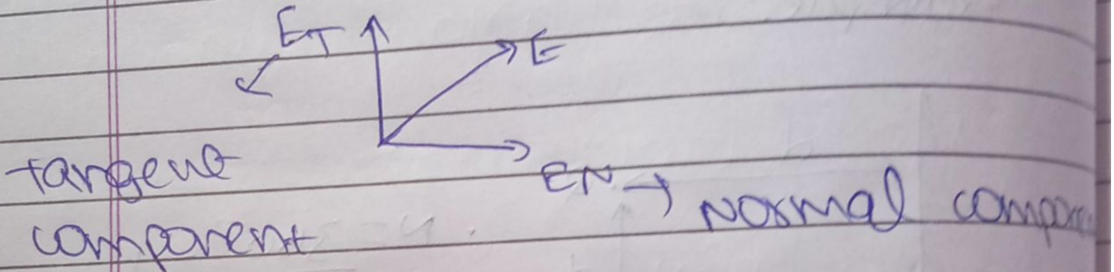
$$N = \frac{360^\circ}{90} - 1 = 3$$

$$\boxed{N = 3}$$

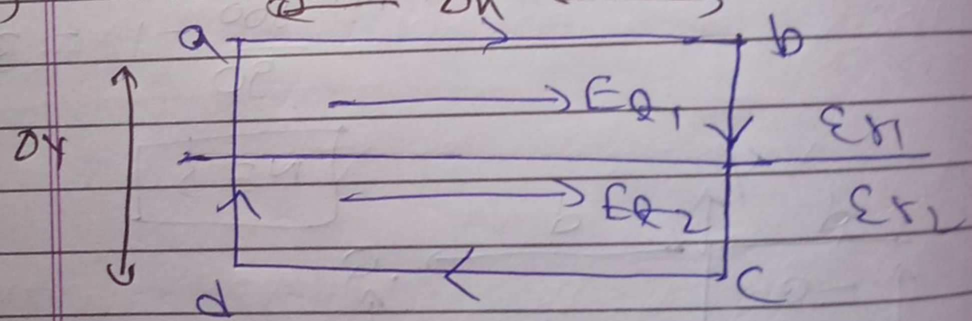


Boundary condition for electric field





(ii) Tangent components



from maxwell's 2nd eqn

$$\oint \vec{E} \cdot d\vec{l} = 0.$$

$$\int_a^b E\,dl + \int_b^c E\,dl + \int_c^d E\,dl + \int_d^a E\,dl = 0$$

$$E_{\ell_1} \Delta x + 0 - E_{\ell_2} \Delta x + 0 = 0$$

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$$\Delta x\,(E_{t_1} - E_{t_2}) = 0.$$

$$\boxed{E_{t_1} = E_{t_2}.}$$

Electric field is continuous for Tangential component.

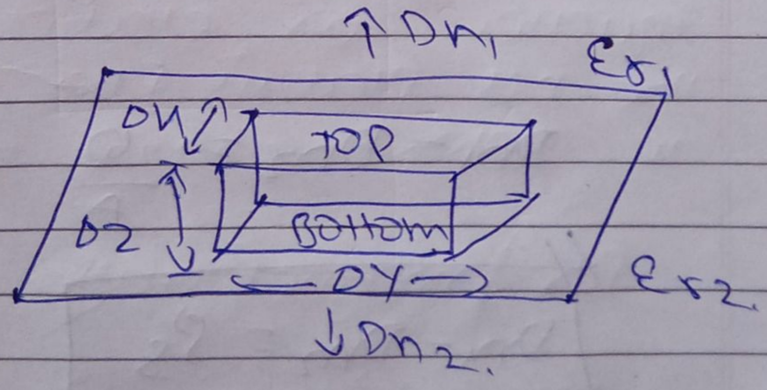
Now,

$$\vec{D} = \varepsilon_r \varepsilon_0 \vec{E}$$

$$\frac{D_{t_1}}{\varepsilon_{r_1}} = \frac{D_{t_2}}{\varepsilon_{r_2}}$$

$$\boxed{\frac{D_{t_1}}{D_{t_2}} = \frac{\varepsilon_{r_1}}{\varepsilon_{r_2}}}$$

(ii) Normal components.



From maxwell's Ist equation

$$\oint \vec{E} \cdot \vec{ds} = \frac{Q}{\varepsilon_0}$$

$$\oint \vec{D} \cdot \vec{ds} = Q$$

$$\int_{top} \vec{D} \cdot \vec{ds} + \int_{bottom} \vec{D} \cdot \vec{ds} + \int_{left} D \cdot ds + \int_{Right} D \cdot ds$$

$$+ \int_{back} D \cdot ds + \int_{point} D \cdot ds = Q$$

$$\overset{\rightarrow top}{D_{n_1}(\Delta y \Delta x)} - \overset{\rightarrow bottom}{D_{n_2}(\Delta y \Delta x)} = Q$$

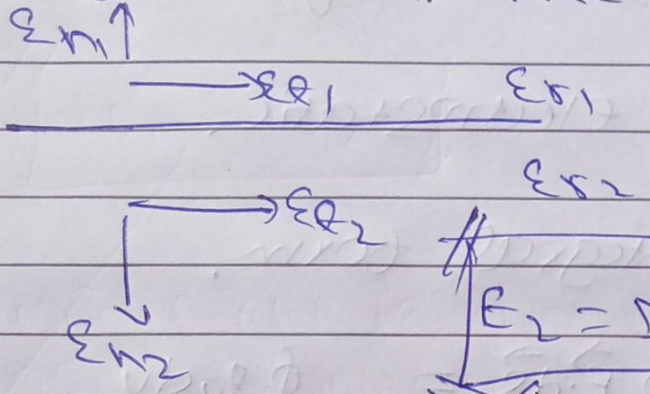
$$\Delta y \Delta x (D_{n_1} - D_{n_2}) = Q$$

$$D_{n_1} - D_{n_2} = \frac{Q}{\Delta y \Delta x} \quad \{ \text{Area} \}$$

$$D_{n_1} - D_{n_2} = \frac{Q}{A} \quad \{ \rho_s = \frac{Q}{A}$$

$$\boxed{D_{n_1} - D_{n_2} = \rho_s}$$

For conductor in free space.



$$\boxed{E_2 = D_2 = 0}$$

$$\boxed{E_{t_1} = E_{t_2} = 0}$$

$$\boxed{\begin{aligned} D_{n_1} &= \rho_s \\ E_{n_1} &= \frac{\rho_s}{\varepsilon_0} \end{aligned}}$$